Sealed Evaluation of Structural World Models: OOD Generalization, Open-Loop Rollout, and Learned-Routing Criterion Validation

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Abstract

World-model errors can preserve scalar loss while violating structural dependencies needed by a decision. Three preregistered exact-state studies first evaluated structural OOD, open-loop rollout, and task-level criterion association. Correct routing separated from random and wrong routing in the first two studies, but a four-arm criterion estimand did not identify routing correctness because a wrong-routed arm also exceeded its point-effect threshold. We therefore ran the source freeze, fresh seed generation, and one-shot execution of a new prospective stochastic study in that order in one scripted sequence. Its 120 independent worlds cross 30 two-edge graph motifs with three transition-head families; two families are outside the original linear TSI equation. Structure and head family are selected from 800 noisy primitive transitions and 400 selection transitions, then evaluated on 1,200 noisier two-coordinate compositions and 160 six-step rollouts per world. Learned graph and head recovery was 1.000, with a multiplicity-adjusted Wilson lower bound of 0.941. Relative to shifted routing, learned routing improved composition negative log likelihood by 0.35933 and open-loop normalized Hamming area by 0.08996. A development-fitted, frozen calibration of discrepancies from the first three transitions predicted terminal failure; correct rather than shifted routing improved Brier score by 0.02402, with simultaneous interval [0.01808, 0.02997]. All nine frozen gates passed, including outside-family noninferiority. This cohort, its root seed, and its eight-quantity multiplicity family are shared with Paper 4; the endpoints reported here are a subset of that single frozen analysis, not independent evidence. A separate Node.js implementation reproduced all 120 learned fits and ten analysis means. Thus the new evidence resolves routing correctness in this declared stochastic family; it does not convert the earlier pooled association into that result.

1 Introduction

Joint-embedding predictive architectures and structured world models seek representations that support prediction and planning without reconstructing every observed coordinate [1, 2, 6]. A scalar latent loss, however, does not identify which part of a structured state failed. Two predictions with the same latent error may differ sharply in whether they preserve a typed label, an edge, a metric comparison, a directed relation, or an order constraint. This matters when utility depends on one of those predicates rather than on average latent fidelity.

This paper studies a narrower question than general world-model superiority: under a declared finite structural signature, can structural discrepancies detect errors that survive matched scalar baselines, and do those discrepancies remain informative under independent world generation, open-loop composition, and plan selection? The mathematical interface uses a single typed correspondence across five layers. Correspondence-based distances are established tools for comparing metric and structured objects [7, 10]; our contribution here is not the general idea of correspondence. It is the frozen combination of a coherent multi-layer state, codebook-free constructive decoding, routing-specific controls, and failure-preserving sealed evaluation.

The empirical study has four design commitments.

(i) Development and sealed worlds are generated from disjoint committed seeds, and each sealed gate is executed once.

(ii) Six controls share input information, active-parameter count, optimization updates, and tuning budget.

(iii) Worlds, rather than transitions or tasks, are the independent inferential units; optimizer seeds are nested replicates.

(iv) Positive results and failed boundaries are reported together.

Three gates are evaluated. The first tests structural out-of-distribution (OOD) prediction under unseen mechanisms. The second tests recursive open-loop error over horizons up to 32. The third asks whether task-local structural discrepancies improve a frozen probabilistic model of plan failure. Proper Brier scores provide the probabilistic criterion [4], and a discrete-time hazard represents first-failure timing [8]. Predictor development uses leave-one-world-out assessment and is frozen before sealed evaluation, following the general separation between model development and evaluation data [3, 9].

The third gate passes its prespecified primary analysis, but its interpretation is intentionally limited. Its discrepancies and regret labels share the same held-out candidate trajectories; this is a same-task criterion association, not a deployment-time prognosis that is available without oracle endpoints. Moreover, a stronger raw layer-aware sensitivity does not reach the prespecified effect floor, and two temporally separated development diagnostics fail. These negative results prevent the scalar-baseline result from being read as evidence that the full correspondence metric is superior to every layerwise structural audit.

2 Frozen Structural Interface

2.1 Coherent states

Fix a finite typed schema and a positive signature Θ. A state has the form

$$\Sigma = (F, \lambda, K, d, \preceq; \Theta) \in \mathsf{CState}(\Theta),$$

where F presents finite typed relations, λ gives tagged labels, K is a simplicial complex on the tagged carrier, d is an intrinsic metric, and $\preceq$ is a finite preorder. Declared cross-layer bridges are membership conditions for $\mathsf{CState}(\Theta)$, not optional penalties.

For states Σ and Γ, a typed correspondence $C \in \mathsf{Corr}(\Sigma, \Gamma)$ is a family of type-preserving relations whose two coordinate projections are surjective. The same C is used to compute label, simplicial, normalized metric, relational, and order distortions, denoted

$$\delta_\lambda(C), \quad \delta_K(C), \quad \delta_d(C), \quad \delta_R(C), \quad \delta_P(C).$$

With positive weights fixed by Θ, the quotient discrepancy is

$$\Delta_\Theta(\Sigma, \Gamma) = \min_{C \in \mathsf{Corr}(\Sigma,\Gamma)} \left(w_\lambda \delta_\lambda(C) + w_K \delta_K(C) + w_d \delta_d(C) + w_R \delta_R(C) + w_P \delta_P(C) \right).$$

The companion theory proves that this is a pseudometric before quotienting and a metric on structural-isomorphism classes. The empirical code computes an exact minimizing correspondence on the finite benchmark.

2.2 Fixed-carrier audit

The benchmark also retains stable typed identifiers. Under that fixed alignment it records a vector

$$e_{\text{fix}} = (e_\lambda, e_K, e_d, e_R, e_P)$$

and its signature-weighted total. This audit is stronger than quotient equality: it can distinguish two isomorphic states whose presented identifiers do not agree. Quotient and fixed-carrier quantities are reported separately.

2.3 Constructive decoding and routing

Every model emits layerwise categorical decisions that are converted to valid states by shared constructive operators: closure for topology, metric closure, declared relational channels, and reflexive-transitive order closure. The primary decoder never projects to an enumerated list of held-out target states.

The six controls are

(1) a matched active dense model;

(2) a layer-routed dense-action model;

(3) a strict factorized-action model;

(4) an oracle signature-routed model;

(5) a random-routed model with matched sparsity; and

(6) a permuted or wrong-routed model.

They receive the same observable fields and use 420 active parameters and the same optimization budget. Oracle signature routing is an upper-bound control, not learned structure discovery.

2.4 Task outcome

Each criterion-validity task supplies two candidate action plans. An exogenous utility checks one primary and one secondary structural predicate, with weights two and one. If $\widehat{j}$ is the plan selected from model predictions and $j^\star$ is oracle-best, regret is

$$\mathcal{R} = U(\Sigma_{j^\star}) - U(\Sigma_{\widehat{j}}), \qquad Y = \mathbf{1}\{\mathcal{R} > 0\}.$$

No correspondence discrepancy, fixed-layer loss, or fitted predictor enters the generator, utility, regret, or failure label.

3 Sealed Multi-Gate Design

3.1 Cohorts and one-shot access

Table 1 summarizes the three confirmatory cohorts. Each gate has a development generator, a separate encrypted seed commitment, a zero-access audit, a frozen source digest, and an atomic one-shot lock. A sealed result is not reused to change a threshold, feature, model population, or sample size.

Table 1: Confirmatory cohorts. Runs equal worlds times six controls times three optimizer seeds.

Gate	Worlds	Runs	Max horizon	Primary purpose
Structural OOD	50	900	1	unseen mechanisms and routing
Open-loop rollout	62	1,116	32	recursive structural error
Criterion validity	69	1,242	8	plan failure and first failure

The OOD and rollout gates use all six controls in paired comparisons. The criterion-validity gate fits its hazard model on the four controls that produce nondegenerate errors. The dense and oracle-signature controls remain mandatory structural-zero checks and are never deleted from the report.

3.2 Structural OOD gate

Development and sealed mechanisms are disjoint at the world level. The generator distinguishes unseen recombinations, unseen structural modes, unseen mechanism parameters, unseen action compositions, bridge-consistent shifts, and bridge-violating negative controls. The primary comparisons pair correct routing against random and wrong routing while preserving capacity and compute. Fifty sealed worlds were selected by the frozen power rule.

3.3 Open-loop gate

The rollout protocol separates teacher-forced and recursive predictions at horizons $1, 2, 4, 8, 16, 32$. It records the area under the Δ_Θ trajectory, the fixed-layer vector, survival without structural failure, bridge violation, and tracking diagnostics. The recursive inequality audit is retained even when its accumulated worst-case numerical bound is vacuous.

3.4 Criterion-validity gate

Each world contains 32 units. A unit has a domain-separated eight-step probe and eight tasks with horizons

$$(1, 2, 4, 8, 1, 2, 4, 8).$$

Task utilities are generated only from structural predicates. The primary population contains the layer-routed, strict-factorized, random-routed, and wrong-routed controls.

The scalar baseline includes training negative log likelihood, one-step and open-loop latent mean-squared error, endpoint exactness, task horizon and goal weights, model identity, plan-selection margin, and selected-versus-alternative scalar error contrasts. The augmented model adds prespecified task-local Δ_Θ, fixed-total, fixed-layer, and goal-aligned structural contrasts. A mandatory non-gating sensitivity gives the baseline direct raw layer mismatches.

A ridge logistic discrete hazard $h_i(t)$ is fit on development worlds. Its cumulative failure probability is

$$\widehat{F}_i(t) = 1 - \prod_{s=1}^{t} \{1 - \widehat{h}_i(s)\}.$$

All coefficients, feature order, means, scales, and penalties are frozen before the sealed seed is revealed.

3.5 Estimands and inference

The two co-primary effects are

$$\mathcal{E}_{\text{bin}} = \text{BS}_{\text{scalar}}(8) - \text{BS}_{\text{TSI}}(8),$$
$$\mathcal{E}_{\text{int}} = \frac{1}{8}\sum_{t=1}^{8} \left[\text{BS}_{\text{scalar}}(t) - \text{BS}_{\text{TSI}}(t)\right].$$

Here, BS(8) is the Brier score at terminal time index $t = 8$ of the fixed eight-step candidate trajectory; the numeral is a time index, not a task identifier. The integrated estimand averages the eight time indices of that trajectory. Positive values favor the structural augmentation. Scores are first averaged within world over units, primary controls, and nested optimizer seeds. One-sided world-level Student tests use Holm familywise correction [5]. We also require positive Bonferroni simultaneous lower bounds from a 20,000-replicate world-cluster bootstrap and from a world-random-intercept, seed-nested variance decomposition. Both point effects must satisfy the prespecified smallest effect size of interest $\varepsilon_{\min} = 0.0025$. The inferential null remains a nonpositive effect; we do not claim that a confidence lower bound exceeds $\varepsilon_{\min}$.

4 Structural OOD Results

All 900 planned fits completed without numerical failure. The codebook-free decoder produced valid structural states, and the paired information, parameter, update, and tuning audits passed.

On the primary sealed structural-OOD endpoint, the oracle signature-routed and matched dense exact controls had zero error. Relative to the signature-routed control, matched random routing and permuted or wrong routing produced the same paired structural effect to machine precision, 0.1943489362, with the same world SD 0.0011631. The two controls are therefore distinguishable from oracle routing in this cohort but not from each other. The dense noninferiority margin was the deterministic constant 0.05 up to floating-point error (world SD 2.10×10^{-17}). The frozen Holm, world-cluster bootstrap, and hierarchical world/seed decisions were positive, but their extreme p-values describe low-variance synthetic margins rather than independent evidence of broad generalization.

This result identifies the value of the declared routing mask in the finite oracle-signature regime. It does not show that the mask can be discovered from data. The unseen mechanism-parameter slice retained a signature-routing error of 0.11995; it is a preserved failure boundary rather than an exception removed from the aggregate.

5 Open-Loop Rollout Results

The rollout gate completed all 1,116 fits on 62 fresh sealed context worlds. The oracle signature-routed and matched dense controls had zero 32-step open-loop Δ_Θ area and unit structural survival.

Against the signature control, the random-routed and wrong-routed controls had mean open-loop area effects 0.17457 and 0.17490, respectively. The frozen family contained eight co-primary margins. Six were deterministic structural margins with effectively zero between-world SD: 0.10, 0.15, and four instances of 0.05. Only the random- and wrong-routing comparisons carried material stochastic variation, with between-world SDs 0.01196 and 0.01120. All eight recorded decisions were positive after multiplicity, cluster-bootstrap, and hierarchical processing, but this is not eight independent informative tests; the stochastic content lies in the two routing comparisons.

Tracking error was zero for every control. This is a useful consistency check but provides no identification of routing quality in a fixed-cardinality world. The exact recursive inequality also passed its finite audit, yet the global worst-case bound for failed controls grew to approximately 1.24×10^{13}. The bound is therefore proof-valid but numerically vacuous in this regime. Neither observation is converted into a positive tracking or stability claim.

6 Task-Level Criterion-Association Results

6.1 Development freeze

The first development specification failed before any sealed reveal: leave-one-world-out improvements were 0.000421 for the terminal Brier score and -0.000221 for the integrated score. The effect floor was not lowered. The revised specification changed the exogenous task utility from mode equality to structural predicates, used task-varying hazards and scalar selected-versus-alternative contrasts, doubled units per world from 16 to 32, and retained the stronger layer-aware sensitivity.

Across 24 revised development worlds, the primary event rate was 0.60384. Leave-one-world-out terminal and integrated improvements were 0.003124 and 0.003927. An analytic floor and a 20,000-draw conjunctive Holm power simulation selected 69 sealed worlds; simulated conjunctive power was 0.9208, with Monte Carlo 95% lower bound 0.91698.

6.2 Sealed primary analysis

All 1,242 sealed fits completed. The primary population contained $26{,}496 = 69 \times 32 \times 3 \times 4$ unit records (worlds, units, nested optimizer seeds, and primary arms), with event rate 0.61632; both exact controls retained event rate zero. Table 2 gives the co-primary analysis.

Table 2: Frozen co-primary criterion-validity results across 69 worlds. Bootstrap and hierarchical columns are simultaneous lower bounds against zero.

Effect	Mean	World SD	Holm p	Bootstrap LB	Hierarchical LB
Terminal Brier	0.003519	0.009015	0.001230	0.001428	0.001392
Integrated first failure	0.003034	0.007471	0.001230	0.001296	0.001271

Both means exceed $\varepsilon_{\min} = 0.0025$. Mean terminal Brier score decreased from 0.183661 to 0.180143, and integrated Brier score decreased from 0.148317 to 0.145283. Outcome noncircularity, panel completeness, event balance, no-refitting, exact-control structural zero, Holm, cluster-bootstrap, and hierarchical requirements all passed.

The pooled estimand averages four heterogeneous arms. Table 3 therefore reports the prespecified secondary arm decomposition before any routing interpretation.

The wrong-routed arm exceeds $\varepsilon_{\min}$ on both outcomes, and the random-routed arm exceeds it on the integrated outcome. Consequently, the pooled criterion analysis identifies an association

Table 3: Arm-level Brier improvements over the frozen scalar model. These are secondary descriptive effects, not separately multiplicity-gated estimands.

Arm	Terminal	Integrated
Layer-routed dense	0.005402	0.003222
Strict factorized	0.003114	0.003077
Random-routed matched sparsity	0.002445	0.002758
Permuted or wrong-routed	0.003114	0.003077

between structural augmentation and the same-task outcomes, not the effect of routing correctness. The larger descriptive effects for layer-routed dense than for strict factorization also do not isolate factorization as the active component.

6.3 Sensitivity and temporal boundaries

When direct raw layer mismatches were added to the generic baseline, structural augmentation improved terminal and integrated Brier scores by 0.002108 and 0.001467. These effects are positive but below $\varepsilon_{\min}$. Thus the primary result establishes an increment beyond a scalar generic baseline, not a practically nontrivial advantage beyond the prespecified layer-aware baseline.

Two stronger temporal interpretations were tested only on development data and failed. Using the independent probe to predict later task failures yielded -0.000539 and -0.000212. Using the first four tasks as calibration to predict the later four yielded -0.002021 and -0.001069. Consequently, the sealed positive result is a same-task criterion association using oracle candidate endpoints. It is not evidence for future-task prognosis or an oracle-free deployment score.

7 Prospective Routing-Resolution Study

7.1 Frozen design

The follow-up estimand and source were fixed before a fresh 256-bit root seed was generated. The freeze, seed generation, and one-shot execution ran in that order in one scripted sequence; integrity follows from the frozen hashes and seed commitment, not from elapsed delay. Each of 120 independent worlds has five coordinates modulo 7 and one graph motif consisting of two distinct sources entering a common target. The 30 possible motifs are crossed with edge heads

$$a_s(1 + x_t), \qquad a_s(1 + x_t)^2, \qquad a_s(1 + x_s + x_t).$$

One third of worlds use two original linear heads; two thirds use at least one of the other heads. Direct multipliers and two edge coefficients are fitted from 800 primitive-action transitions with coordinate noise 0.08. Graph and head family are selected using 400 disjoint transitions. No fitting reads the 1,200 OOD transitions, all of which have two active action coordinates and coordinate noise 0.12. The designated composition stratum activates both causal edges simultaneously.

The criterion comparison uses the same learned predictor under two routing maps. During the first three steps of a six-step stochastic trajectory, the correct score records target-coordinate discrepancies when either learned source is active; the negative control routes the same discrepancy field through a fixed shifted graph. A logistic map fitted on 24 public development worlds was frozen before confirmation. Its outcome is terminal mismatch at the learned target, and its estimand is wrong-routing minus correct-routing Brier score. Thus the outcome does not contain either criterion score, the diagnostic precedes the terminal outcome, and the two arms differ only in routing.

Eight inferential quantities use Bonferroni simultaneous two-sided intervals with familywise level 0.05. The identification gate additionally requires a simultaneous Wilson lower bound of 0.90. Point-effect thresholds are 0.04 for the learned composition NLL contrast, 0.025 for rollout Hamming area, and 0.005 for criterion Brier score. These point thresholds are distinct from the null boundary used by the simultaneous intervals.

The analysis object reports ten interval-bearing quantities at the frozen Bonferroni divisor of eight. They contain one exact duplicate pair, one sign-flipped pair, and one deterministic zero, leaving seven informationally distinct stochastic quantities. The frozen v1 contract specified only the integer divisor and did not enumerate its members; this is a reporting defect, not a preregistered family definition. A post-review sensitivity using divisor ten leaves every decision unchanged. The cohort, root seed, and multiplicity family are shared with Paper 4; the three endpoints reported here are a subset of that single frozen analysis, not independent evidence.

No power analysis for 120 worlds was frozen before confirmation. A post-review retrospective design justification, using only the 24 development worlds and 20,000 stratified bootstrap draws, estimates conjunctive nine-gate power at 0.9694 with Monte Carlo interval [0.9670, 0.9718]. This calculation does not use confirmatory results and cannot retroactively establish a preregistered sample-size rationale.

7.2 Confirmatory results

The learned motif and both head families were recovered in all 120 worlds (rate 1.000; simultaneous Wilson lower bound 0.941). Table 4 reports the three Paper 3 endpoints. Every interval is computed across worlds; cases and rollouts are nested observations.

Table 4: Prospective learned-routing effects. Positive values favor learned or correct routing.

Endpoint	Mean	World SD	Simultaneous interval
Composition NLL	0.35933	0.02012	[0.35431, 0.36435]
Rollout Hamming area	0.08996	0.01162	[0.08706, 0.09286]
Routing-correct Brier	0.02402	0.02381	[0.01808, 0.02997]

All three point effects exceed their frozen thresholds and all simultaneous lower bounds exceed zero. Performance on the 80 worlds outside the original linear family was noninferior to the 40 linear worlds: outside-minus-linear NLL was -0.00132 with simultaneous interval [-0.01031, 0.00766], whose upper bound is below the 0.03 margin. This is a direct learned-routing criterion result, not a reinterpretation of the earlier four-arm average.

In a separate post-review descriptive sensitivity, train noise $\{0.04, 0.08, 0.16\}$ was crossed with OOD noise $\{0.06, 0.12, 0.24\}$, with 120 public-seed worlds per cell. Graph and head recovery remained 1.000 in all nine cells. The minimum mean learned-routing NLL effect was 0.23337. This panel was not confirmatory and varies only the declared coordinate-corruption process.

8 Scope and Limitations

Structural discovery scope. The promoted structure result uses observation-only algorithmic graph discovery from primitive transitions followed by factorized parameter recovery. It does not establish that an unconstrained neural network discovers the same graph. The neural structure track remains exploratory.

Synthetic and bounded-cardinality regime. All confirmatory worlds come from declared finite synthetic families. The variable-cardinality audit covers three finite cardinality panels, but births, deaths, and held-out entity identity turnover are absent. The zero transition error therefore does not test identity maintenance under turnover.

Criterion rather than temporal validity. The task-local discrepancy and regret outcome use the same held-out candidate trajectories. TSI does not enter the outcome definition, but oracle endpoints are required to compute both the generic error comparators and structural discrepancies. The failed temporally separated diagnostics rule out a stronger prognostic interpretation in the present benchmark.

The original co-primary criterion estimand averages four arms, including random and wrong routing. Because wrong routing exceeds the point-effect threshold on both outcomes, that analysis is not a routing-correctness test. The prospective study resolves this particular ambiguity with a direct correct-versus-shifted routing estimand; it does not alter the estimand or interpretation of the historical cohort.

Comparator dependence. The co-primary improvement is relative to a scalar generic model. The layer-aware sensitivity remains positive but below the effect floor. The current evidence therefore does not identify the common-correspondence optimization itself as superior to all simpler layerwise structural descriptions.

Magnitude and uncertainty. The Brier improvements are small. Their simultaneous lower bounds are positive, but they do not exceed the 0.0025 point-effect threshold. The analysis supports a positive effect plus a point estimate above the threshold; it does not establish that the population effect exceeds that threshold with 95% confidence.

External validity and noise boundary. The prospective result covers stochastic coordinate corruption, three head families, learned routing, and held-out two-mechanism composition in a finite state space. A project-independent Node.js code path reproduces all 120 learned factorized fits and the ten reported analysis means. This is an independent implementation check, not an independent research-group study. Visual inputs, entity turnover, and real-world systems are outside the defined population and are not inferred from these results.

9 Reproducibility and Artifact Integrity

The revision bundle contains the implementation, generators, frozen predictors, analysis plans, access ledgers, raw results, reports, and the final scope audit under package-relative `src/`, `experiments/`, and `reproduction/` paths. No local machine path is needed to locate them. The criterion-validity access ledger records one seed reveal and one result evaluation.

The freeze value

```
40d3d2083601f1214e2f26fc9833d22e97d369e3c6471646a3b5390124dc972f
```

is an internally computed combined digest declared in `artifact_gate.json` and the access ledger; it is not the SHA-256 of either containing file. The earlier manifest, raw-result, analysis, and report digest strings were likewise internal payload digests. We therefore do not use those self-declared fields as an independent integrity check.

Instead, direct SHA-256 hashes of the transferred files are:

manifest:
`0ec0b96685828d84432216a5bc7e10ddccd6f293963f6975117d3d698523c189`
raw result:
`3d6c69af546e311fb15c0c8e521e776eb79a9ba88dc6389e5457c45325b4e734`
confirmatory analysis:
`2cb260a084a6c6175e6b5312bcb32a6dcadafd16035f6c2123abc08c0fe3da0b`
evidence report:
`43d5643c95ecb5cb9268d2f57ccfce19214dc18392e31f399ddd993b01816211`
access ledger:
`8c8651cef755a2a3c8a9be5bf3ddc3a86af382721369cc72902c783fb8fe144b`
scope audit:
`f6151ca50dfdbe50e23558b6f18a9325300bae8c695d52df4145a3d3a6110ab9`.

The six historical files named above are included in the package-relative `experiments/` tree; the reviewer-response README maps each label to its file. The bundle-level `SHA256SUMS` independently covers every transferred file. The final scope audit is retained as a historical, superseded artifact and is not used to assign a current grade.

No cumulative "evidence level" is assigned. Within this paper, completed obligations are limited to the sealed OOD, sealed rollout, same-task criterion, failure-preservation, and artifact-integrity checks described above. Separate supporting artifacts examine algorithmic graph discovery, bounded noisy perception, variable cardinality, a packaged benchmark runner, and a second finite mechanism family, but they do not convert this study into learned neural structure discovery, public-domain validation, or independent replication.

The packaged runner is `PYTHONPATH=src python3 tools/run_paper3_public_benchmark.py`. Two byte-identical legacy outputs establish deterministic same-environment replay only. The prospective extension adds a separate Node.js implementation with zero project imports. It verified the revealed seed commitment and reproduced 120/120 learned graph–head fits and 10/10 analysis means. Direct SHA-256 values for its freeze, raw result, analysis, integrity ledger, portable input, clean-room audit, and JavaScript source are, respectively, `bfdc91ec3156`, `cd89c1ff6055`, `2e6880df0543`, `e0ca28d93c7c`, `3f438a1cfb7d`, `8371860c58c8`, and `b648562bc3b8`. This path consumes exported world data and does not re-derive worlds from the committed seed; generator-side errors and world-seed derivation are outside its coverage. A separate Python derivation audit uses the archived frozen generator to derive all 120 seeds from the revealed root, regenerate every train, selection, and test case, and match the portable export without discrepancy. This verifies seed/export consistency within the frozen Python implementation; it is not a second-language generator replication. English/Korean section structure, mathematical environments, decimal literals, and rendered-text regressions are checked together.

10 Conclusion

The original exact-state studies established sealed OOD and open-loop routing effects and a narrower four-arm same-task criterion association. Their arm-level decomposition correctly showed that the pooled criterion estimand could not identify routing correctness.

The prospective stochastic study resolves that identification problem rather than relabeling the old result. Routing and head family were learned from disjoint noisy training and selection partitions. Correct learned routing then improved composition NLL, six-step rollout error, and temporally ordered criterion Brier score over a shifted route on 120 fresh worlds. The result persists outside the original linear transition equation, and a clean-room Node.js implementation reproduces the learned fits and analysis means.

The resulting claim is exact: within the declared finite stochastic family, train-only structural identification recovers the active two-edge mechanism, and routing that mechanism correctly has predictive and criterion value. The study makes no claim about visual perception, unconstrained neural discovery, or real-world systems because those are different estimands, not unresolved conditions attached to the present conclusion.

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